Algebraic Geometry Princeton Quals Questions

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1 Groups

Question 1.0. (Kollar) *Compute the genus of the curve* $x^3 = y^6 + 1$

Question 1.1. (Katz) *Define the genus of a curve in every way that you know how.*

Question 1.2. (Katz) *Why are these equal? State Serre Duality. Define the Hilbert Polynomial.*

Question 1.3. (Katz) *Compute the genus of the Fermat curve in every way you can.*

Question 1.4. (Katz) *What does adjunction say in the case of the Fermat curve? How about a degree* d *hypersurface?.*

Question 1.5. (Katz) *What are such hypersurfaces called (*$K3$ *surfaces)?*

Question 1.6. (Katz) *What happens if I know the number of points of the variety over a finite field with* p^r *elements for all* r*?*

Question 1.7. (Katz) *Suppose a variety has exactly* $p^n + 1$ *points over a finite field of characteristic* p^n*. Specifically, consider the projective* n*-space,* P^n*, over a finite field with* p^n *elements.*

What can we deduce about this variety?

Do you know how to prove this?

Question 1.8. (Katz) *State the Weil Conjectures*

Question 1.9. (Katz) *Suppose that my equations are defined over $\mathbb{Z}$. What can I say when I base change to various fields?*

Question 1.10. (Katz) *State Riemann-Roch. What is Riemann Roch good for?*

Question 1.11. (Katz) *What's the relationship between divisors and sections of invertible sheaves?*

Question 1.12. (Katz) *What would Weierstrass say about elliptic curves?*

Question 1.13. (Katz) *Define the Weierstrass p-function. What are its basic properties? What can you say about its poles? Does it satisfy a differential equation?*

How would you know that there should exist such a differential equation?

Question 1.14. (Katz) *Give me a 30 second sketch of everything you know about Riemann-Roch in higher dimensions. What is this useful for?*

Question 1.15. (Bhatt) *Embed a genus 1 curve in projective space.*

Question 1.16. (Venkatesh) *Why is it called an elliptic curve?*

Question 1.17. (Bhatt) *What if it's over $\mathbb{R}$ and has no $\mathbb{R}$-points? Then can you embed in $\mathbb{P}^2$? What about a finite field?*

Question 1.18. (Bhatt) *How can you always find an $\mathbb{F}_p$ point of a genus 1 curve?*

Question 1.19. (Bhatt) *Can I embed a genus 1 curve in higher P^N as a complete intersection?*

Question 1.20. (Bhatt) *Give an example of a smooth affine curve which has nontrivial Picard group.*

Question 1.21. (Bhatt) *Can* $\mathrm{Pic}(X)$ *be finitely generated? Work over $\mathbb{C}$ and countable field cases.*

Question 1.22. (Bhatt)

Question 1.23. (Bhatt) *If I blow up X surface at a point how does the cohomology change?*

Question 1.24. (Bhatt)

Question 1.25. (Bhatt) *1*

Question 1.26. (Shimura) *What is a scheme?*

Question 1.27. (Shimura) *Define the genus of a curve (in a couple of ways). Explain how to compute it using the Hilbert polynomial. Why is this the same as the dimension of the space of global 1-forms?*

Question 1.28. (Shimura) *or a coherent sheaf F on P^r associated to the graded module M, why is the Hilbert polynomial H(n) (:= Euler characteristic of F(n)) equal to the dimension of M_n for n >> 0?*

Question 1.29. (Shimura) *Compute the genus of the curve y^2 = x^5 - x by using Hurwitz' formula.*

Question 1.30. (Xu, Chengyang) *What do you know about minimal model program on rational surfaces. Why are they different? How do you show they are connected by blow-up and blow-down?*

Question 1.31. (Xu, Chengyang) *What is the class of the canonical divisor on F_n?*

Question 1.32. (Xu, Chengyang) *In general, consider a ruled surface f: X = P(E) to B on a curve B of higher genus, E is a rank 2 vector bundle, what is the class of the canonical divisor.*

Question 1.33. (Xu, Chengyang) *What is the class group of a smooth affine conic curve (say, in characteristic 0).*

Question 1.34. (Xu, Chengyang)

Question 1.35. (Xu, Chengyang)